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Designing with Evolutive Colors: A Proof of Concept for Bio-based Dye Integration in Textile Design Digital Workflow

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Current industrial textile production relies on petrochemical dyes. This dependence has led to growing interest in bio-based alternatives, although their adoption remains constrained by technical and cultural factors. In particular, bio-based dyes do not meet typical industrial standards for stability, diversity of colour choices, and reproducibility. Textiles dyed with bio-based colourants tend to exhibit colour changes under light exposure. Current design software systems were built with the conventional expectation of stability of petrochemical dyes in mind, making them ill-adapted to bio-based dye behaviour and thus hampering their industrial adoption. To address this gap, we propose the prototype of a novel design software for textile designers, which integrates visualisation of light discoloration of bio-based dyes over time. This proof-of-concept software integrates the characteristics of materials inherently unstable to light in the design of a knitted jacquard. We argue that integrating the specificities of bio-based dyes into design tools could enable designers to transition toward more regenerative design approaches. Grounded in an understanding of the chemical behaviour of dye molecules over time and the translation of these dynamics into computational representations and textile structures, the software encourages broader reflection on the role of digital design tools in influencing designers' choices. This work opens perspectives for the integration of bio-based active materials into contemporary design, production, and use cycles, by reframing instability and evolution as intrinsic properties of the material rather than limitations.

Keywords: natural dyes; predictive modelling; evolutive design; active materials.

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1. Introduction

The contemporary textile industry relies heavily on petrochemical dyes, which originate from non-renewable fossil carbon resources. In 2012, an estimated 20% of water pollution in China was attributed to the dyeing industry (Kant, 2012). Dye fixation rates on fibres typically range from 75% to 95% and, when effluents are not adequately treated, the unfixed portion can be released into the environment as a pollutant (Dieval & Fafet, 2021). Even when these pollutants are removed from wastewater, the treatment process often generates sludge contaminated with chemicals and heavy metals that remain difficult to manage (Rahman, 2019). In response, certain bio-based dyes, colourants derived from renewable biological resources, including plants, fungi, insects, algae, or microorganisms, have been seen as a way of reducing the industry's dependence on these feedstocks. They offer promising alternatives, their adoption is however hindered by several economic, technical, and cultural factors.

Among these factors, lower fastness compared to synthetic dyes represents a significant limitation, as most bio-based dyes exhibit reduced resistance to light exposure, friction, washing, and detergent use (Islam et al., 2024). Materials that diverge from the performance standards established by synthetic dyes are often perceived as lower in quality : textiles that fade tend to be discarded contributing to unsustainable consumer practices (Cooper et al., 2016). This perception hinders the integration of bio-based dyes into industrial design and production systems, which are shaped by standardised quality requirements and a prevailing paradigm of "design for permanence" (Bréchet, 2013). Addressing this gap requires a shift toward approaches that engage with materials as active, complex, and evolving systems (Fratzl et al., 2021; De Visscher, 2023). Such a material-centred perspective has traditionally been more accessible at the artisanal scale, where designer and maker are often the same (Tiragallo, 2015). In industrial contexts, however, designers are rarely involved in the dyeing process and therefore lack direct experiential knowledge of the behaviour of these materials.

More broadly, sociotechnical systems have developed in close alignment with the properties of petrochemical dyes. Contemporary design tools and methodologies, particularly computer-aided design (CAD) systems, operate on the assumption that colours and designs remain unchanged throughout an object's lifecycle. This manifests implicitly in user interfaces of software : the chromatic appearance is chosen as an immutable property while temporal evolution remains an externality (Morovic, 2019). As a result, there is a lack of a systematic approach to visualise, anticipate, and even conceptualise the evolution of colour over time for textile designers. On the other hand, this absence of visualisation implies for the broader public a difficulty to consider temporal discoloration as an acceptable product behaviour (Chapman, 2005).

Chromatic design tools must be rethought (Calvo-Ivanovic, 2025). Access to structured data has been shown to support more sustainable textile design practices (Kapsali & Hall, 2023), suggesting that predictive and visualisation capabilities could enable designers to better engage with material dynamics. Other fields, such as architecture, have already integrated temporal properties and material behaviour into CAD systems, particularly through energy, environmental, and behavioural simulations that account for a building's performance over time. Empirical studies on architectural design processes (e.g. Wang et al., 2021; Gath-Morad et al., 2023) show that access to predictive information about future states can influence design decisions and outcomes. Therefore, the proposed prototype shows how visualising colour transformations over time can support decision-making, particularly in

the selection and combination of bio-based dyes whose behaviours differ and evolve. More broadly, the work highlights that embedding material variability and complexity into design interfaces not only affects design outcomes, but also questions designers' methods and approaches to materials, encouraging more adaptive practices.

2. Background: Technical and Cultural Barriers to the Adoption of Bio-based Dyes

2.1. Bio-based Dyes as Active and Evolving Materials

Alternatives to petrochemical dyes encompass a wide range of sources. Pre-industrial practices drew on plant, animal, and mineral-based colourants to dye textiles. Dyes are soluble colourants binding to the substrate, making them well-fitted for fibre colouration. Before the beginning of the transition to synthetic dyes in 1856 with Perkin's mauveine, certain sources of colour were identified as superior in quality, particularly in terms of fastness, leading to the large-scale cultivation and trade of specific dye materials such as madder, indigo, cochineal or weld (Cardon, 2014). Given the environmental impacts of synthetic dyes and their reliance on petrochemical resources, the introduction of bio-based dyes into industrial markets has been increasingly discussed within the design community over recent decades. In 2008, the Raw Colors studio in the Netherlands raised the issue of introducing plant-based dyes into contemporary industry, textile designer Clément Bottier proposed a small collection produced using natural dyes on a local scale on industrial looms in 2014, and brands such as Patagonia (Clean Color Collection, 2019) have experimented with applications on a larger industrial scale. Some recent approaches of particular interest have tried to consider the shift from petrochemical dyes as a way to increase biodiversity and tackle the issue of invasive plant species (Keskitalo et al., 2023), to valorize industrial by-products (Baaka et al., 2024) and to question modern aesthetics (Niinimaki, 2023). Design projects range from Studio Sarmite's reuse of dyes extracted from discarded textiles to Agnė Kučerenkaitė's experimental project *Ignorance is Bliss*, which investigates by-product materials from a mid-sized Lithuanian farm, and resulted in a series of linen rugs produced using residues of mustard and primrose cultivation. Atelier LUMA, in the south of France, have developed dyeing processes based on invasive species for the tapestry *Untitled* (2021) by Rirkrit Tiravanija. Contemporary developments have broadened prospects, including research on use of algae (Mutaf-Kılıc et al., 2023), fungi (Venil et al., 2020) or bacteria (Gomes, Soares, 2023) as sources of dye production. Developments of bacterial dyes or industrial by-products dyes have been explored and developed by companies like PILI, Colorifix, Green-ing or Ficus.

However, these initiatives have struggled to gain widespread adoption. Among the key factors cited is the lack of colourfastness of these bio-based dyes compared to industrial standards. Dyed materials undergo temporal evolution driven by discolouration kinetics. Discolouration is used here as a broader term than fading, as colour change may involve not only a loss of chroma but also shifts in hue or lightness. Studies of discolouration kinetics allow to anticipate how colour change is likely to occur (Giles, 1965). Kinetics have been characterised across several dye types in the 1980s, revealing, in some cases, substantial colour shifts (Cox Crews, 1987). ISO 105-B02 standard rates lightfastness on a scale of 1 to 8 by comparison with reference dyes. Plant-based dyes rarely exceed a fastness rating of 5 and are more commonly rated below 4, reducing their competitiveness relative to most synthetic dyes on the industrial market, which can request minimum scores between 5 and 6 (Islam et al., 2024). The evolving material properties of common bio-based dyes have long been documented by heritage

conservation research. Since the second half of the twentieth century, research has focused on preventing degradation and discolouration in the preservation and exhibition of cultural heritage artefacts (Cox Crews, 1987; Crews, 1982; Whitmore et al., 1999; Groeneveld et al., 2023). Although research on bacterial dyes is comparatively recent, early studies indicate that they face similar issues of colour instability, highlighting that temporal evolution remains a broader challenge across bio-based dye systems (Gomes & Soares, 2023).

2.2. Limitations of CAD and Digital Design Tools

The influence of design software on creative methods and the forms it produces has been studied since the emergence of computer-aided design in the late twentieth century. In 2003, Matthew Fuller examined the reciprocal relationship between software and users, and particularly how these shape our worldview, beliefs, and actions. Within the field of design, graphic designers have been the first to take up this subject. Lev Manovich 2013's book Software Takes Command, a reference to Mechanization Takes Command (Giedion, 1947) explored how software products such as Photoshop or Illustrator shaped contemporary aesthetics. Alternatives to the conceptual frameworks provided by graphic design software have also been proposed (Blanc, 2023).

In recent years, textile design software has undergone substantial transformation, driven by ongoing technological advancements and the rapid development of digital fashion design tools. In particular, the integration of 3D modelling into design workflows has profoundly reshaped design practices, notably through the emergence of platforms such as CLO3D, Optitex, and Style3D (Eren, 2025). While the introduction of 3D technology for clothing design dates back to the early 1990s (Hinds and McCartney et al. 1990; Hinds et al. 1992), the first 3D algorithmic models for CAD that enabled the transition from a 2D model to a 3D model were introduced in 2003 (Chiricota, 2003; Fuhrmann et al., 2003). Possible solutions to the difficulty to work from 3D to 2D (tri- to two-dimensional) were proposed in the 2015's (Sayem, Kennon, Clarke 2010). Since the mid-2010s, the historically dominant workflow based on 2D design software such as Adobe Illustrator or Lectra, combined with physical prototyping and 2D colour arrangement tools has progressively been replaced by 3D modelling and, more recently, AI-driven generation, accelerating the development of digital tools for textile design.

These include the creation of digital twins of existing textiles through scanning technologies, as developed by companies such as Vizoo; the production of photorealistic 3D textures for visualisation purposes through tools such as Adobe Substance 3D; and the ongoing development of rendering capabilities within textile-specific design software, which remain only partially operational and rarely exportable for 3D. Among those experiments, knitted visualisations have placed particular emphasis on colour representation, partly due to the challenges of accurately rendering the complexity of knitted structures and bindings digitally. While simple jacquard patterns can be visually simulated through texture mapping, more complex weaving and knitting structures demand computational approaches capable of translating the logic of textile construction into digital representations. Existing design software systems emphasize visual rendering at the point of production, treating colour as a fixed attribute. This limitation becomes particularly critical in multicoloured patterns like jacquard fabrics, where differing discolouration kinetics between dyes can lead to unexpected visual effects.

2.3. Research questions

Building on these considerations, we consider how the evolving nature of bio-based dyes can be integrated into design processes. Rather than treating instability as a defect to be corrected, we ask how it might be approached as an intrinsic material property to design with. This perspective raises several questions: what does it mean to design with colours that transform over time? How can the temporal evolution of colour effectively be visualised within design tools? What types of data are required to support such representations, and how can these temporal dynamics be communicated to designers and users? More broadly this research aims to start an investigation on how the integration of evolving material parameters might challenge current methods in textile design.

3. Design and Operation of a Tool for Design with Bio-Based Dyes

To explore these research questions we propose a software prototype which takes in consideration the evolution of colour properties in time. An initial prototype was developed based on well-established web-based frameworks (next.js, three.js). This tool was developed as a simple web-based interface to explore how designers could appropriate this tool. A choice was made to not provide any backend services (e.g. server-side computations, or data-storage), allowing this tool to be more easily tested from individuals and organisations without data ownership and confidentiality concerns.

This CAD tool was conceived as a proof of concept based on a knitted jacquard workflow, chosen to limit the number of structural textile variables and isolate the influence of colour-related parameters. It aims to operate between the graphic design phase and physical prototyping. More specifically, it is positioned within the 2D/3D previsualisation stage that has emerged in digital textile design workflows, enabling a reduction in physical prototyping. Below we present the rationale and functionalities of the main components of this prototype.

3.1. Colorimetric evolution

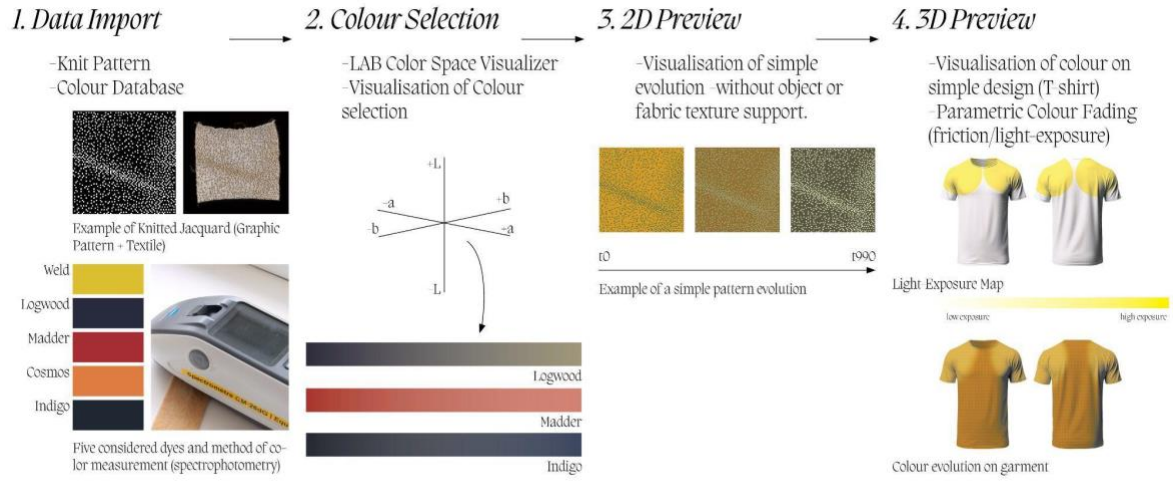


Figure 1. Architecture of the prototype.

Our prototype integrates simulated data of colorimetric evolution of natural dyes under light, for the design of a jacquard-knitted t-shirt up to six colours. The interface is divided into four steps (Fig. 1): data import (colorimetric and pattern), colour selection assistance, 2D preview, and 3D preview taking into account uneven fading on a three-dimensional object. The workflow is linear, as each stage was developed independently.

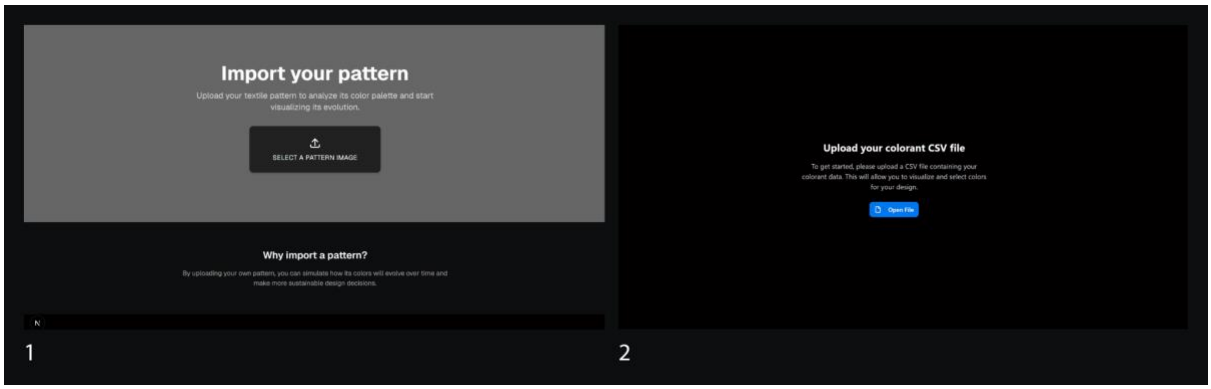


Figure 2. Screenshot from the prototype : (1) Import of pattern and recognition of colour areas (2) Import of colour data LAB as CSV file.



Figure 3. (1) Bi-colour (2) Tricolour (3) Quadri-colour pattern, 24 pixels by 24 pixels designed with Adobe Photoshop loaded in the prototype

The first step of the prototype enables users to import custom patterns and datasets. This allows designers to bring their own data. The input pattern must be provided as an image containing up to six colours as the standard flatbed knitting machine is of eight yarn guides with two yarn guides being reserved for the cast-on yarn and the joining yarn. Pattern processing treats each image pixel as a stitch, assigning it a corresponding colour to be linked to dye data (Fig. 3). The colour data is then mapped to each detected colour region of the design.

We use as an example a restricted dataset of four well-known natural dyes, applied at 100% owf[1] : indigo (*Indigofera tinctoria*), madder (*Rubia tinctorum*), logwood (*Haematoxylum campechianum)*, weld (*Reseda luteola*), and sulphur cosmos (*Cosmos sulphureus*) on felted wool mordanted with alum 20% owf, before and after 530 hours of artificial exposure to a xenon lamp (Table 1). The evolution trajectories are defined by initial and final colorimetric coordinates in the CIELa*b* space (a tridimensional colorimetric space, where x = a*, y = b*, z = L) and linear interpolations for the intermediate values. The dye dataset is then implemented in the workflow on a timeline ranging from 0 to 990 units, with coordinates sampled every 30 units, resulting in 33 colour values.

[1] *On weight of fibre*

Table 1. Colorimetric coordinates L, a, b* of the dyes used as dataset example, at initial and final exposure time (0 and 530 hours). Intermediate colour values were obtained by linear interpolation.*

	Initial coordinates			**Final coordinates**		
	L	**a***	**b***	**L**	**a***	**b***
Indigo (*Indigofera tinctoria)*	21,5	0,0	-9,6	25,6	-1,0	-10,6
Madder (*Rubia tinctorum*)	34,8	35,3	21,3	54,0	29,0	17,2
Weld (*Reseda luteola*)	71,3	1,8	62,9	78,5	0,3	26,8
Sulphur cosmos (*Cosmos sulphureus*)	54,7	29,2	48,2	79,1	1,2	20,3
Logwood (*Haematoxylum campechianum)*	16,4	3,0	-8,2	51,5	1,4	8,8

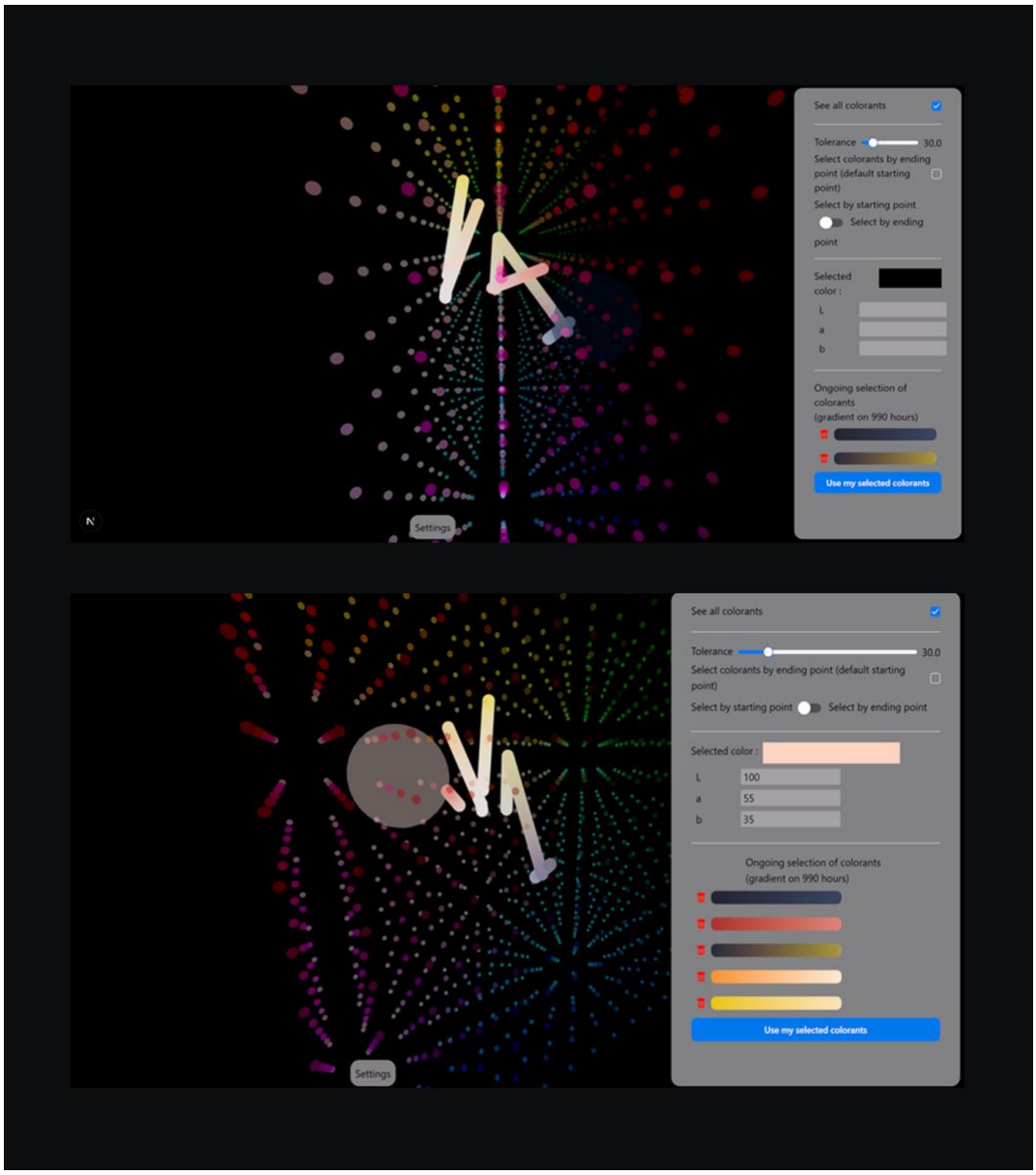


Figure 5. Screenshot from the prototype : colour selection environment of the interface.

Figure 5 shows the second step which supports the designer's colour selection by accounting for the colorimetric constraints they face when designing with bio-based dyes. The evolving colour data is represented as an interactive data visualisation in a 3D colour space . The user can select colours and visualise how they change over time or select a specific colour with L*a*b* coordinates. To account for the constrained and heterogeneous colour gamut of bio-based dyes, the tool incorporates a tolerance range around the initially selected shade. Within this range, it displays alternative achievable colours together with their projected evolution over time. Colour evolution is represented as a polyline gradient from an initial value at t_0 to the end value at t_{990}. This enables designers to identify cases where initial visually similar shades derived from different dye sources exhibit distinct colourfastness behaviours, and to make informed decisions based not only on the initial appearance but also on future colour changes.

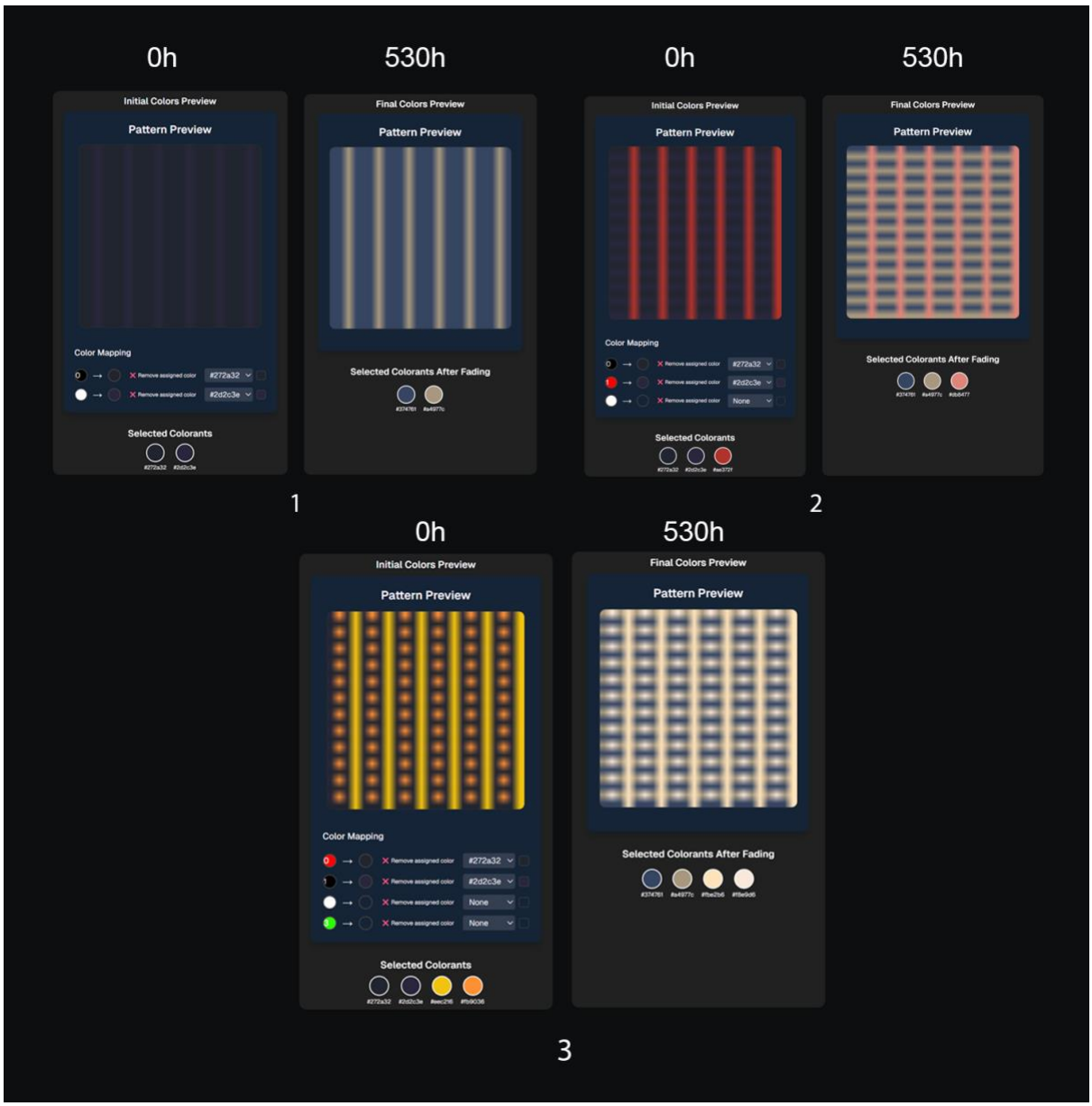


Figure 6. Screenshot from the prototype : (1) Bicolour indigo and logwood dye (2) Tricolour indigo, logwood, madder dye (3) Quadricolour pattern indigo, logwood, weld, sulphur cosmos dye.

In the third step, the user can visualise the chosen colours as a 2D visualisation. Two complementary functionalities are proposed: (a) the visualisation of individual colour evolution, and (b) the pixelated pattern visualisation under its initial and final colour schemes (Fig. 6). This step allows the designer to experiment with several aesthetic variations based on different dyes and to visualise how they evolve over time. Recolouring is here extended to include temporal evolution: the design can be evaluated not only at the moment of conception but also as it transforms over time. This enables designers to identify and select colour combinations that preserve the visual balance of the pattern over time.

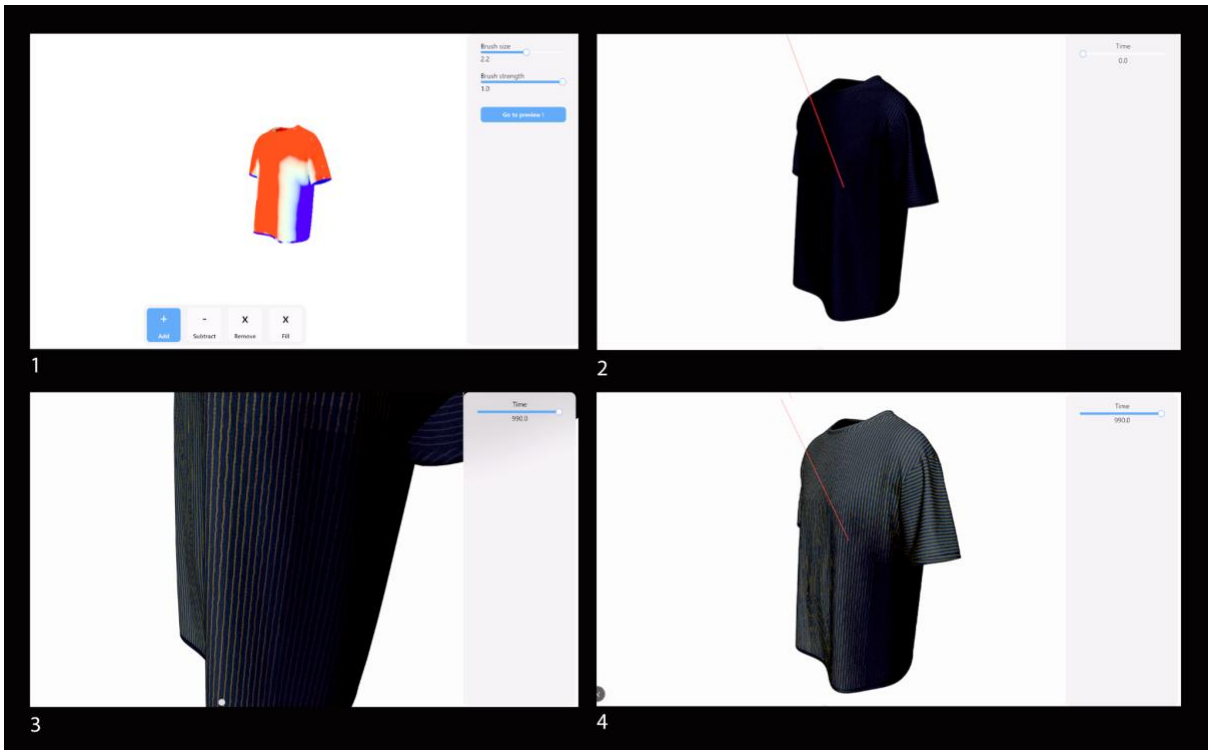


Figure 7. Screenshot from the prototype : (1) Heatmap on T-shirt (2) T-shirt dyed with logwood and indigo before discolouration (3) close-up over the textile visualisation at the end of the lifecycle of the object, white lines pattern appear on the parts dyed with logwood (4) T-shirt after partial degradation considering the heatmap.

At a fourth and final step, the tool offers a 3D visualisation of the pattern applied to a previously loaded 3D model of the object or garment (using the common .obj file format). This can be combined with user-defined areas most exposed to light fading, represented as “heatmaps” (Fig. 7.1). For instance, in the case of a T-shirt, exposure is unevenly distributed: shoulders are more exposed to light, whereas underarms remain relatively protected, depending on the garment’s volume and use conditions. The software therefore enables the integration of usage scenarios into the design process, supporting the anticipation and mapping of degradation patterns over time. After configuring the light-fading areas, the garment is displayed in a time-based visualisation that simulates how the selected colour schemes evolve over time under uneven exposure conditions.

4. Discussion

The primary contribution of this paper lies in the integration of colour ageing models into the design process, through a novel interactive design tool enabling the visualisation of temporal colour evolution.

4.1. Functional scope and constraints of the prototype

At its initial version, this prototype is particularly effective in 2D representations (Step 3), where colour changes can be accurately rendered and compared. In contrast, initial exploration in 3D visualisation (step 4) highlights challenges: lighting conditions of the 3D model reduces colour accuracy, making these representations less reliable for colour decision-making at this stage. However, compared to 2D visualisation, 3D enables more realistic projections of use conditions. Currently, the prototype supports a single 3D use scenario, limiting its flexibility and scope of application. The selection of a T-shirt as a constrained scenario allows for the manual generation of heatmaps to simulate exposure conditions. While this approach offers flexibility and encourages designers to critically consider how different areas of an object are used and exposed, it is time-consuming for the designer and limits scalability to other types of products.

4.2. Beyond static colour, integrating temporal parameters and material behaviour in a design tool

Incorporating the degradation of dyes under the influence of light into the visualisation phase of the design process represents a significant departure from what current software takes into account. The tool must account for the time-dependent evolution of material properties, adopting a long-term perspective on the object. It encourages designers to consider temporality in their design decisions, including the successive colour states the textile undergoes. For instance, indigo is relatively lightfast, whereas logwood is sensitive to light, resulting in markedly different fading behaviours (Fig. 6.1). When combined with madder (Fig. 6.2), additional chromatic shifts emerge over time: the red of madder gradually fades toward lighter pink tones, potentially altering the balance of the pattern and requiring a reevaluation by the designer. In conventional textile design software, this stage of visualisation corresponds to the colour variation or recolouring phase, where alternative colourways are generated for a given pattern using tools such as Pointcarré or NEDRecolor (Nedgraphics). However, these tools operate within a static framework, focusing on the appearance of colour at the point of production and market presentation, without accounting for its potential evolution during use. The prototyped tool here situates the textile object within intertwined temporal, use, and material frameworks. It positions dye as an active component of the design process and could encourage reflection on the object's lifecycle. This contrasts with conventional textile software approaches, in which colour is treated primarily as a visual attribute, detached from its material and physicochemical properties.

4.3. Designing under conditions of material and data variability in design workflows

Plant-based colourants are typically classified according to their chemical composition (Yusuf et al., 2017), encompassing categories such as quinones, flavonoids, anthocyanins, tannins, indigoids, carotenoids, curcuminoids, and berberines (Cardon & Du Chatenet, 1990). Unlike petrochemical dyes,

they are often complex mixtures rather than single molecules. For example, madder contains approximately fourteen chromophoric compounds (Cardon, 2014). While it is possible to approximate initial colour and general fading tendencies as shown in the dataset, precise predictions remain limited, even for diluted composition, as the relative proportions of the diverse molecules vary. With regard to fading predictions, the sensitivity of these dyes to contextual factors such as light exposure, humidity, temperature, and conditions of use introduces further variability that can influence their evolution and reduce the reliability of predictive models. The integration of user-defined datasets allows for the incorporation of local and situated knowledge, while simultaneously raising questions regarding data reliability, accessibility, and interpretation. Rather than resolving these tensions, the tool makes them visible, opening the possibility for future developments that could incorporate probabilistic visualisations, confidence or prediction intervals, or alternative modes of representing uncertainty.

This research highlights the role of data in the design process. In doing so, it interrogates designers' tolerance for material change, challenging conventional design approaches founded on assumptions of material stability. As the tool addresses the issue of stability in time, it does not address the degree of uncertainty of such predictions (de Visscher, 2023) which imply that designers will have to engage with ranges of variation between projected and realised colour states, forcing a reconsideration of notions of control and quality in relation to material behaviour. In doing so, it makes visible a question that remains largely absent from current CAD tools: how can these tools help designers to engage with uncertainty when the future behaviour of materials cannot be predicted with complete accuracy? In contrast to tendencies toward standardisation and simplification, this experiment confronts us with complex material realities that do not readily conform to established regimes of measurement and exactitude (Bensaude-Vincent, 2024). It opens discussions on the role the designer will have in navigating this variability and complexity (Thackara, 2006), maybe shifting the role of the designer from controlling a stable outcome to engaging with a range of transformations. This repositioning might invite new design approaches in which variations are not only mitigated but can be anticipated, composed, and integrated as part of the aesthetic and functional qualities of the textile object.

4.4. Future work and perspectives

The production of light-sensitive designs has already been explored by designers and researchers in design. In 2020, Dutch designer Nicky Vollebregt, through *Uncover*, developed the idea of combining natural and petrochemical dyes with differing fastness levels to produce patterns. In the same year, *Design to Fade* was realised for PUMA by the Dutch project Living Colour and the Swedish design studio Streamateria, using bacterial dyes. Since 2021, it has constituted a central axis of designer and researcher Lou Ramage's practice (Ramage, 2025). The idea of mixing dyes fastness was further developed by the American Studio Meyers in their project *Fades Made to Fade* in 2024. By challenging our perception of wear or discolouration, these projects suggest an interest for such designs. To evaluate the relevance of the prototype, testing with both designers specialised in such approaches and more conventional textile designers will be necessary to assess needs, usability, and potential areas for improvement. It will also help clarify how such a tool may influence design and production practices, and whether it can contribute to overcoming cultural barriers to the adoption of bio-based dyes. Indeed, the acceptance of evolving textiles such as denim in specific cultural contexts indicates that attitudes toward colour change can be malleable (Lilley et al., 2019).

Among the limitations due to this early-stage development is the linear structure of the prototype, which does not reflect the iterative and comparative nature of design practice. Designing with colour involves the need to produce and compare multiple variations simultaneously, enabling connections between stages and supporting non-linear exploration would constitute an important direction for future development. At present, the absence of established colourfastness databases for bio or even plant-based dyes restricts the range of colours available in the prototype, highlighting the need for further work on validated, shared datasets as well as predictive modelling approaches. Tolerance in relation to data accuracy and uncertainty ranges associated with expected colours must be addressed prior to testing with professional designers. This will require the development of appropriate visualisation features that take into account ranges of possible pattern outcomes in order to help the evaluation of acceptable design results. From an industrial applicability perspective, additional factors such as friction and wash fastness would need to be incorporated, taking into account similar challenges related to data variability as those identified in the present study.

5. Conclusion

This research highlights how design tools can influence not only the forms produced, but also the ways in which design practices engage with materials. By introducing colour variability over time as a design parameter, the prototype explores how digital tools may support the consideration of ageing, uncertainty, and transformation within textile design processes. Rather than replacing approaches based on stability, it suggests possibilities for integrating material change as an additional dimension of design exploration. The material, here colour, is repositioned as a potential design parameter rather than a fixed attribute. This experimentation raises questions regarding the formats and directions of future developments in 3D textile visualisation tools. There is great interest in better understanding the possible agency of such software in shaping the cultural acceptance of material dynamics, and whether these approaches could open pathways toward more adaptive, and materially grounded design practices.

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